Norwegian Subsea

Accurate Motion Measurements in Real Sea Conditions

Motion Reference Units

High performance, Affordable and Robust



MRU Compact



MRU Marine



MRU Subsea

Norwegian Subsea

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Updated Aug 2026

Our advantages

High Accuracy

An MRU from Norwegian Subsea provides robust and high-performance motion measurements even during irregular motion and horizontal accelerations.

Other MRUs may perform well in predictable motion, such as a single frequency sine wave motion in lab conditions, but a Norwegian Subsea MRU gives you the performance you need in real sea conditions.

Affordable

High performance in all conditions at affordable prices set the Norwegian Subsea MRUs apart from the competition.

Easy to use

Our small and lightweight MRUs, with flexible cable options, are easy to install and come with free web browser configuration software for effortless integration into your system.

How do we achieve this?

MEMS sensors
Advances in MEMS technology gyroscopes and accelerometers enable new high accuracy sensors at low cost

+

Sensor Fusion Algorithms
Our unique advanced sensor fusion algorithms leverage the best available sensors to provide precision results under realistic circumstances

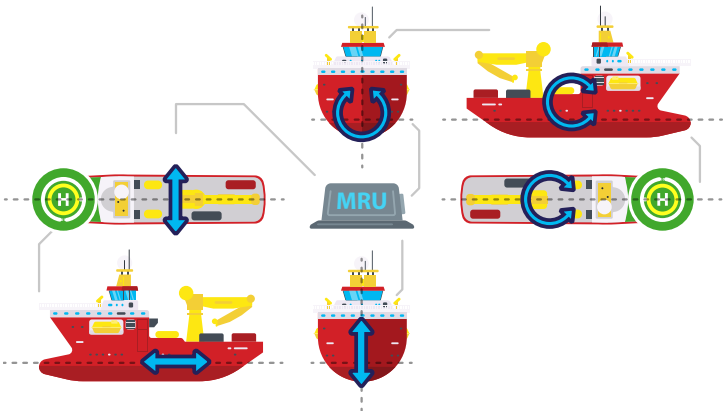
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High Accuracy at Low Cost
High Accuracy in Real Sea Conditions

What is a Motion Reference Unit?

A Motion Reference Unit (MRU) is a self-contained sensor that measures motion in all 6 degrees of freedom (DoF): Roll, Pitch, Yaw, Surge, Sway, and Heave. The 6 DoF velocities and accelerations are measured by the MRU using high-end gyroscopes and accelerometers together with advanced sensor fusion algorithms.

An MRU has high accuracy Roll & Pitch measurements, and measures oscillatory Heave, Surge and Sway motions for wave periods at sea. Linear motions with very long periods, or steps, cannot be measured by an MRU as it assumes a 0 mean heave position.



Product overview

The Norwegian Subsea MRU range covers most marine and subsea applications.

The MRU Compact, Marine, and Subsea each feature mechanical and electrical interfaces tailored to different environments, ensuring a suitable option whether installed in a cabinet, on deck, or on the seafloor.

Each model is available in 3000, 6000 and 9000 series variants, to accommodate different accuracy requirements. All MRU models are also available in Vertical Reference Unit (VRU) versions for applications that only require roll and pitch measurements. All Norwegian Subsea MRUs can be mounted in any orientation.

MRU Compact



Usage area
IP 65

Heave accuracy
5.0 cm or 5.0 %

Roll & Pitch accuracy

3000 ± 0.05°
6000 ± 0.02°
9000 ± 0.01°

Connectors
RJ45 or RJ50

MRU Marine



Usage area
IP 68, 50 m depth

Heave accuracy
5.0 cm or 5.0 %

Roll & Pitch accuracy

3000 ± 0.05°
6000 ± 0.02°
9000 ± 0.01°

Connectors
Lemo 16-wire, Subconn 8-wire or Subconn 16-wire

MRU Subsea



Usage area
6000 m depth

Heave accuracy
5.0 cm or 5.0 %

Roll & Pitch accuracy

3000 ± 0.05°
6000 ± 0.02°
9000 ± 0.01°

Connectors
Subconn 8-wire

MRU Ex



Usage area
Hazardous areas

Heave accuracy
5.0 cm or 5.0 %

Roll & Pitch accuracy

3000 ± 0.05°
6000 ± 0.02°
9000 ± 0.01°

Cable
Pigtail cable 10-wire or 4-wire

Which series should I choose?

This depends on your application. The following guidelines can help you decide.

3000

The 3000 series, with 0.05° accuracy in Roll & Pitch, is ideal for most ***monitoring applications*** where the main purpose is to collect motion data for analysis or decision making. This model is also suitable for some control applications, such as AHC for smaller vessels with short lever arms to the remote monitoring points.

6000

The 6000 series, with 0.02° accuracy in Roll & Pitch, is our most versatile MRU. Its high accuracy makes it suitable for the majority of use cases and it is the recommended MRU for most ***control applications***, including AHC and winch control.

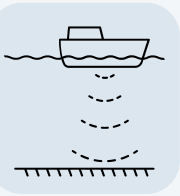
9000

The 9000 series, with 0.01° accuracy in Roll & Pitch is our most accurate MRU. We recommend this MRU for the most demanding sonar motion compensation applications. This model is also recommended if high accuracy surge and sway measurements are important.

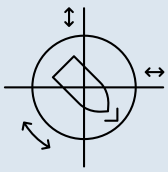
Applications

Norwegian Subsea MRUs can be used in a variety of Marine, Subsea and Wind Energy applications.

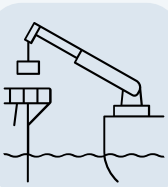
Marine Operations



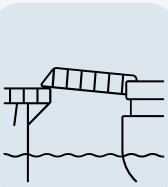
Hydrography
Sonar systems require accurate motion compensation to produce high-quality seafloor maps, even in demanding sea conditions.



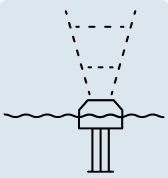
Dynamic Positioning
Easy to retrofit MRU with no recalibration for motion compensation of GNSS antenna and other position reference systems.



Active Heave Compensation
Heave compensation for cranes, LARS and winches requires accurate heave measurements in real sea conditions.



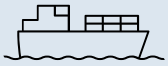
Gangway
Ensuring the safety of 3D motion compensated gangways in real sea conditions requires accurate 6DoF motion measurements in real-time.



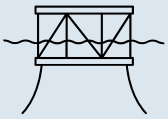
Wind Lidar Buoys
The delivery of high quality wind data requires accurate correction for heave, roll and pitch measurements in even the most extreme conditions.



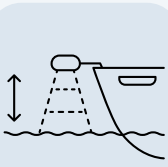
Helideck Monitoring
Accurate heave velocity and roll and pitch measurements in real sea conditions. No recalibration needed and flawless operation in any temperature.



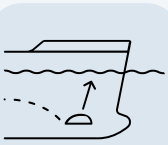
Ship Motion Monitoring
Accurate ship motion data supports vessel performance optimisation and helps prevent cargo damage or container loss.



Offshore Fish Farms
Condition monitoring of Offshore fish farms.
Reduce risk with Accurate motion measurements.

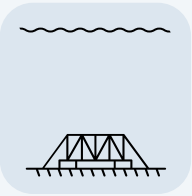


Wave Radar
Reliable wave radar data depends on accurate heave compensation, with compact systems requiring no recalibration to ensure high uptime.

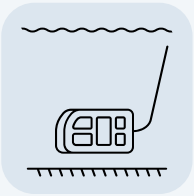


Stabilizing Fins
Roll damping systems and stabilizing fins need accurate and affordable roll measurements in real sea conditions.

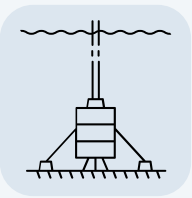
Subsea



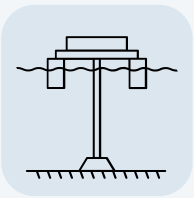
Subsea Surveys
Precise subsea tilt (roll & pitch) measurements at affordable prices.



ROV/AUV
ROV and AUV operations require accurate motion data for reliable navigation and effective payload sensor performance.



BOP Monitoring
Accurate 6DoF motion and vibration measurements to monitor BOP and other subsea structures.

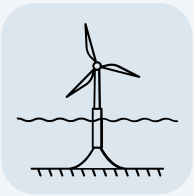


Riser Monitoring
Topside motion sensors as input to the riser digital twin or subsea sensors attached to the riser.

Wind Energy



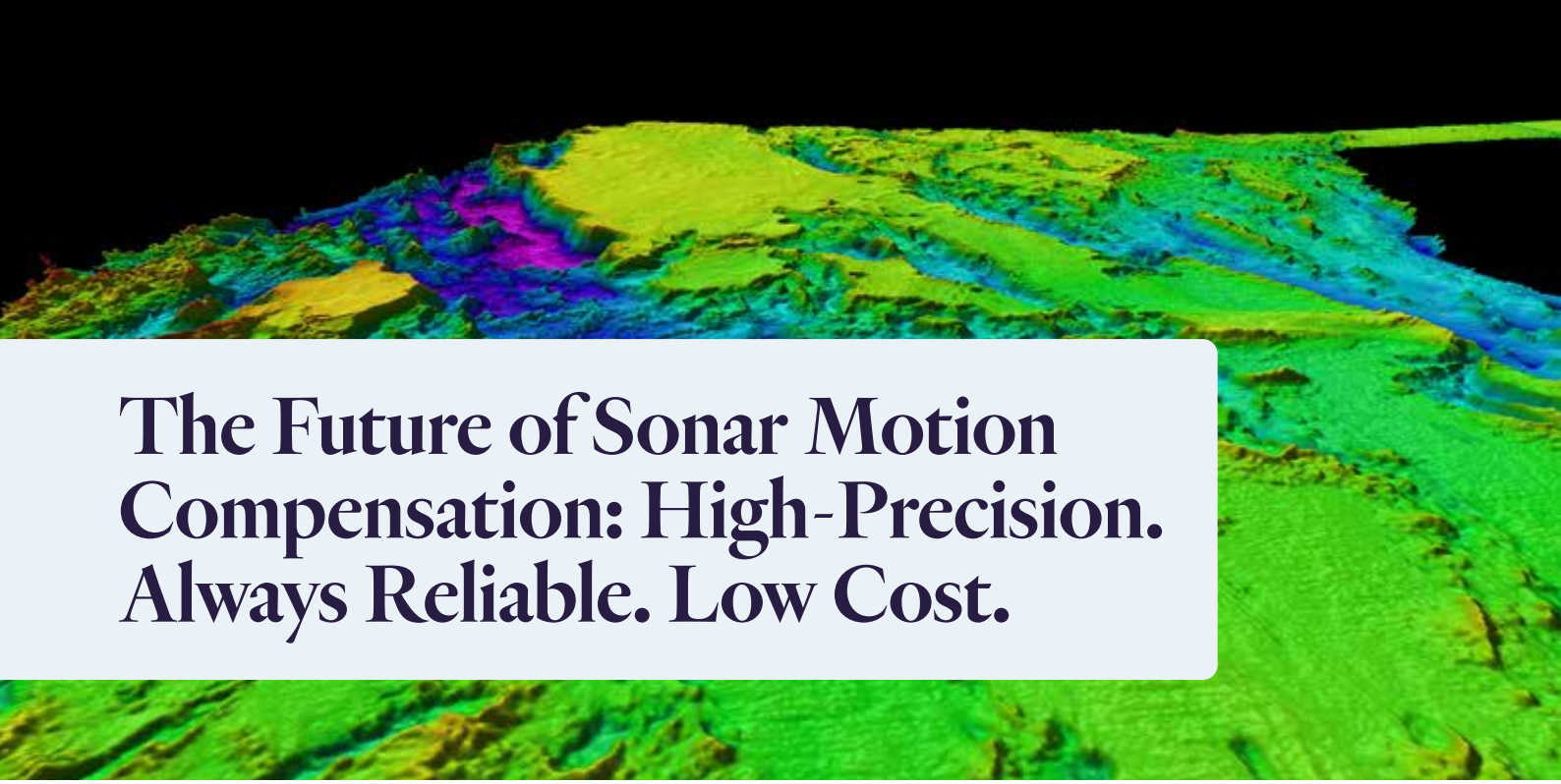
Bottom Fixed Offshore Wind
Turbine tower roll & pitch measurements as input to a wind tower digital twin model.
Ultra-low-noise accelerometer for structural monitoring.



Floating Wind Turbine Control
Accurate MRU measurements used in active pitch control to limit wave induced motions.



Floating Wind Base Motion Monitoring
Structural monitoring of the floating wind base and mooring system using MRUs and Ultra-low-noise accelerometers.



The Future of Sonar Motion Compensation: High-Precision. Always Reliable. Low Cost.

The Game-Changer for Sonar Accuracy in Challenging Environments

MRUs are crucial in bathymetric survey technology, providing precise motion compensation data to ensure reliable and accurate seafloor mapping. The Norwegian Subsea MRU redefines what is possible, delivering exceptional performance even in incredibly challenging conditions. Our technology provides ***high-accuracy roll, pitch, and heave compensation*** without dependency on RTK or GNSS signals, eliminating common vulnerabilities in sonar mapping. All our MRUs meet IHO Class 1 standards for hydrographic accuracy.

Why you need Motion Compensation for Sonar Systems?

- → ***Accurate Seafloor Mapping*** – Without proper motion compensation, vessel movement can distort sonar readings, leading to incorrect depth measurements and unreliable bathymetric data.
- → ***Elimination of Data Gaps*** – Uncompensated motion causes inconsistencies in sonar returns, resulting in gaps or misaligned sonar images, making it harder to detect underwater features.
- → ***Improved Target Detection*** – Whether for shipwreck hunting, fish-finding, or subsea infrastructure monitoring, stable motion data ensures clear, high-resolution sonar images.
- → ***Avoid repeated surveys and extensive post processing*** – High-quality motion compensation ensures high-precision sonar results in real-time.

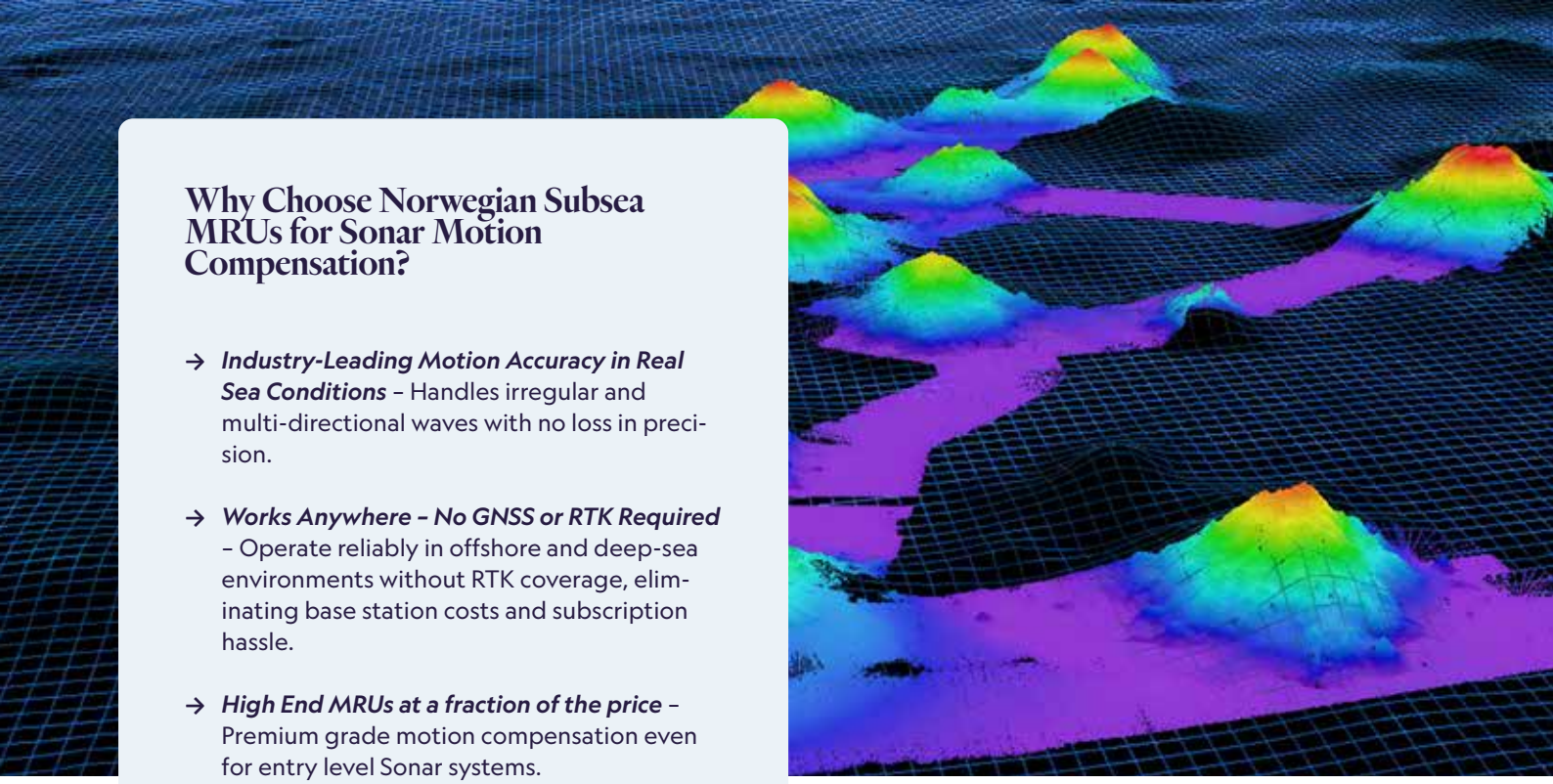
Captain Sid Hynes struggled with traditional motion sensors when mapping shipwrecks in Newfoundland's Grand Banks—a region notorious for unpredictable, three-directional 6 feet swells.

"The standard sensor was jumping all over, creating gaps in the data. But the Norwegian Subsea MRU was like a needle—smooth and precise. You could barely tell the boat was moving. It is incredibly sensitive compared to the others"

– Sid Hynes, Master Mariner



Courtesy Olex AS



Why Choose Norwegian Subsea MRUs for Sonar Motion Compensation?

- → ***Industry-Leading Motion Accuracy in Real Sea Conditions*** – Handles irregular and multi-directional waves with no loss in precision.
- → ***Works Anywhere – No GNSS or RTK Required*** – Operate reliably in offshore and deep-sea environments without RTK coverage, eliminating base station costs and subscription hassle.
- → ***High End MRUs at a fraction of the price*** – Premium grade motion compensation even for entry level Sonar systems.
- → ***Real-Time Heave, Roll, and Pitch Correction*** – Achieve survey-grade data accuracy in all conditions in real-time.
- → ***Ready-Calibrated and Maintenance-Free*** – Reduce downtime and ensure long-term reliability in real sea conditions.
- → ***Compact Form Factor*** – Can be mounted on vessels, near sonar heads, or integrated within sonar equipment.
- → ***Plug-and-Play Integration*** – Compatible with leading sonar systems for seamless operation.
- → ***OEM Solutions Available*** – Can be integrated directly into sonar systems for optimized performance.

"With the Norwegian Subsea MRU, our customers get a system that is easy to use, highly accurate, and doesn't require a massive learning curve. The results speak for themselves."

– Nick Fogarty, Senior Product Manager, WASSP

"The accuracy, weight, and form factor of Norwegian Subsea MRUs are unmatched."

– Gabriel Walton, Technical Director, Ultrabeam Hydrographics

Optimized for all types of sonar systems

Norwegian Subsea MRUs seamlessly integrate with ***Multibeam Echo Sounders (MBES), Side-Scan Sonars (SSS), Single Beam Echo Sounders (SBES), Forward-Looking Sonars (FLS), and Sub-bottom Profilers,*** ensuring superior motion compensation across all sonar applications.

Norwegian Subsea MRUs redefine sonar motion compensation by delivering ***high-performance accuracy at an affordable price.***

Upgrade your sonar system today. All sonar systems can benefit greatly from using Norwegian Subseas high-precision motion compensation.

MRU Compact



Compact, Plug & Play
The eMRU is compact, dust and splash proof (IP 65).
The eMRU is plug & play due to the standard RJ 45 ethernet connector with passive power over ethernet (PoE).
This makes cabling fast & easy. It comes with a wide range of standard and customized data protocols.
Integration is never an issue.

Heave accuracy
5.0 cm or 5.0 %

Available models

3000	6000	9000
± 0.05°	± 0.02°	± 0.01°

Roll & Pitch accuracy

	MRU Compact	*MRU Compact RJ50*
Environment	IP65	
Connector	RJ45 with POE	10-pin RJ50 with POE
Ports	Ethernet	Ethernet & RS-232 / Ethernet & RS-485 / Ethernet & PPS
Communication	Ethernet: UDP, Modbus TCP, Ethernet/IP	
Weight	0.55 kg	
Footprint (L × B x H)	12.0 cm × 5.7 cm × 5.3 cm	
Enclosure material	Anodized aluminum 6061-T6	
Temperature range	Operating: -40 to +85 °C Storage: -40 to +85 °C	

What is passive Power over Ethernet?
Passive Power over Ethernet means that the MRU Compact is powered through the Ethernet cable with 9 – 36 V DC.

We recommend using a PoE splitter / injector to power the MRU Compact.

Versions and Options
1. VRU version without Heave/Surge/Sway output, which gives the same Roll & Pitch performance as the MRU, also in irregular coupled motions.
2. Magnetic Heading

MRU Marine



Versatile Placement
The MRU Marine can be used on deck or underwater at depths up to 50 m (MRU Marine SW), offering a compact and cost-effective solution for mounting close to the sonar head. It includes Ethernet, RS232, RS485, and PPS time synchronization ports for easy integration with your system.

Heave accuracy
5.0 cm or 5.0 %

Available models

3000	6000	9000
± 0.05°	± 0.02°	± 0.01°

Roll & Pitch accuracy

	MRU Marine	*MRU Marine SW / MRU Marine SW16*	
Environment	IP68	50 m water depth	
Connector	Marine 16 pin male connector	8 pin SubConn MCBH8M	16 pin SubConn MCBH16M
Ports	Ethernet, RS-232, RS-485, PPS	Ethernet & RS-232 / Ethernet & 2-wire RS-485	Ethernet, RS-232, RS-485,PPS
Communication	Ethernet: UDP, Modbus TCP, Ethernet/IP. RS-485: Modbus RTU		
Weight	1.2 kg		
Footprint (L × B x H)	15.4 cm × 8.6 cm × 6.7 cm		
Enclosure material	Anodized aluminum 6061-T6		
Temperature range	Operating: -40 to +85 °C Storage: -40 to +85 °C		

Analog outputs
Analog outputs are available using the analog output junction box. Configurable +/-10 V and 4-20 mA output signals are available.

Retrofit
Norwegian Subsea MRUs are often used as a drop-in replacement of older MRUs (retrofit) on Dynamic Positioning, Helideck Monitoring and Active Heave Compensation systems. For these applications, normally no recalibration is required.

Versions and Options
1. VRU version without Heave/Surge/Sway output, which gives the same Roll & Pitch performance as the MRU, also in irregular coupled motions.
2. Magnetic Heading

MRU Subsea



6000 m Water Depth Rating
The MRU Subsea is ideal for subsea applications such as riser motion monitoring, ROV/AUV operations, and subsea surveys. Housed in a compact titanium casing and depth-rated to 6000 m, the unit has a small footprint that allows for easy installation almost anywhere. It also features a Subconn wet-mateable connector.

The Subsea Inclinometer version is popular due to its small size, high accuracy, and affordability.

All Norwegian Subsea sensors can be mounted in any orientation.

Heave accuracy
5.0 cm or 5.0 %

Available models

3000	6000	9000
± 0.05°	± 0.02°	± 0.01°

Roll & Pitch accuracy

	MRU Subsea
Environment	6000 m water depth
Connector	8-pin SubConn FCR1508M
Ports	Ethernet & RS-232 Ethernet & 2-wire RS-485
Communication	Ethernet: UDP, Modbus TCP, Ethernet/IP
Weight	1.6 kg
Footprint (L × B x H)	7.6 cm × 7.6 cm × 16.5 cm
Enclosure material	Titanium housing
Temperature range	Operating: -40 to +70 °C Storage: -40 to +85 °C

Easy interfacing
The MRU Subsea comes in a 6000 m rated titanium housing with ethernet and serial ports for easy communication with your system. Industrial communication protocols can be used for PLC interfacing. The MRU comes with a wide range of standard and customized ASCII or binary data protocols. The MRU can be delivered with custom length cables with Subconn 8 pin connectors.

Versions and Options
1. VRU version without Heave/Surge/Sway output, which gives the same Roll & Pitch performance as the MRU, also in irregular coupled motions.
2. Magnetic Heading

MRU Ex



For Hazardous Areas
The MRU Ex is a cost-effective solution for hazardous areas. It's easy to install with custom pigtail cable lengths and supplied with all required certificates and documentation.

- Ex certificate II 2 G Ex d IIC Gb
- Compact and cost effective unit
- Stainless steel unit
- Ethernet & serial line

Heave accuracy
5.0 cm or 5.0 %

Available models

3000	6000	9000
± 0.05°	± 0.02°	± 0.01°

Roll & Pitch accuracy

	MRU Ex		
Environment	II 2 G Ex d IIC Gb		
Cable	Fixed pigtail cable with custom length		
	10-wire	8-wire	4-wire
Ports	Ethernet & RS-232 / Ethernet & RS-485	4-wire RS-485 / Ethernet & RS-232 / Ethernet & 2-wire RS-485	RS-232 / 2-wire RS-485
Communication	Ethernet: UDP, Modbus TCP, Ethernet/IP. RS-485: Modbus RTU		
Weight	19,5 kg (excluding cable)		
Footprint (L × B x H)	19 cm ×19 cm × 20.5 cm		
Enclosure material	AISI 316L		
Temperature range	Operating: -20 to +60 °C Storage: -40 to +85 °C		

Versions and Options
1. VRU version without Heave/Surge/Sway output, which gives the same Roll & Pitch performance as the MRU, also in irregular coupled motions.

MRU Specifications

	3000	6000	9000	*Remarks*
		Performance		
Roll & Pitch	± 0.05°	± 0.02°	± 0.01°	RMS (dynamic)
Heave (real-time)	5.0 cm or 5.0 %	5.0 cm or 5.0 %	5.0 cm or 5.0 %	Whichever is greater for 0 to 25 s periods
Heading (optional)	± 0.5°	± 0.5°	± 0.5°	Magnetic heading
		Range		
Rotation speed	± 150°/s	± 450°/s	± 450°/s	-
Acceleration	± 3 g	± 4 g	± 10 g	-
Heave	± 50 m	± 50 m	± 50 m	-
Roll/Pitch/Yaw	± 180° / ± 90° / ± 360°			-
Output frequency	0–100 Hz	0–100 Hz	0–100 Hz	Adjustable output frequencies
		Gyro output		
Scale factor error	0.2% max/min	0.2% max/min	0.2% max/min	-
Angular rate noise	0.05°/s RMS	0.025°/s RMS	0.015°/s RMS	-
		Acceleration output		
Acceleration noise	0.0025 m/s^2 RMS	0.002 m/s^2 RMS	0.0015 m/s^2 RMS	-
Acceleration accuracy	0.01 m/s^2 RMS	0.01 m/s^2 RMS	0.01 m/s^2 RMS	-
		Power & Interface		
Power consumption	6 W			-
Supply voltage	9-36 V DC (24 V nominal)			-
Internal storage	32 GB			-
Data protocols	Custom NMEA, ASCII and Binary, Atlas, Gyrocompas 1, Ifremer Victor, MDL, NORSUB 1-8, Simrad EM 3000, SMCA, SMCC, Tokimek PTVG, RDI ADCP, TSS1 and many more. Please refer to user manual.			-
Configuration protocols	GET/SET through telnet or serial port and web server based configuration SW			-
Time synchronization	Sync to host PC, Sync to NTP server, Sync to PPS			-

	3000	6000	9000	*Remarks*
		Features		
PPS		PPS time synchronization		-
Aiding		Heading and Velocity aiding		On request
		Environmental specifications		
Electromagnetic compatibility		IEC 60945/EN 60945 (immunity/emission)		-
Vibration		IEC 60945/EN 60945		-
Max shock		2000 m/s2 (half-sine 0.5 msec) non-operational (10 ms peak)		-
MTBF (computed)		100000 h		-

ACC Marine Ultra-Low-Noise Accelerometer

Built for the signals standard accelerometers can't see.

The ACC Marine is a 3-axis MEMS accelerometer with a noise density of 0.7 µg/√Hz RMS — low enough to resolve structural motion that standard sensors cannot distinguish from noise. Exceptional velocity random walk (VRW) and bias instability make it ideal for demanding vibration monitoring.

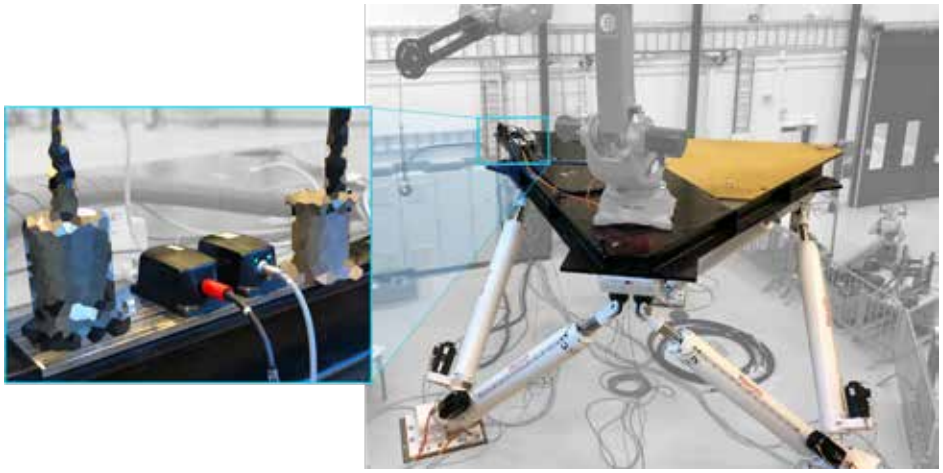
A compact waterproof aluminium housing means no external enclosure and no custom signal conditioning. Mount the ACC Marine in a dry area, or the ACC Marine SW on a floating substructure in up to 50 m water depth. Ethernet and serial ports with Modbus TCP, Modbus RTU, and EtherNet/IP make integration straightforward.

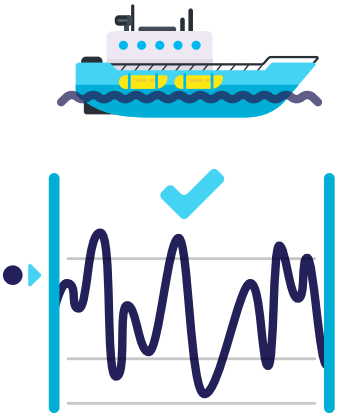
Developed for floating offshore wind platforms, jackets, monopiles, topsides, and vessel hulls — anywhere structural response, resonance behaviour, or wave- and machinery-induced vibration must be measured. The ACC Marine supports real-time integrity monitoring, serves as input to digital twin models, and enables fatigue life analysis.

	ACC Marine
Noise density	***0.7 µg/sqrt(Hz) RMS***
Range	± 15 g
Scale factor	6.00E-05 mg/LSB
Misalignment axis to axis	0.1 degs
Nonlinearity	0.03 %FS
Initial error	2 mg
Bias repeatability	3 mg
In run bias instability	0.2 µg
Velocity random walk (VRW)	0.00012 m/s/sqrt(h)
Frequency bandwidth 3dB	400 Hz

Validated in real sea conditions

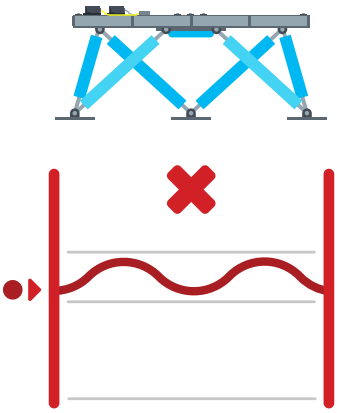
Tested in real sea conditions
Many suppliers tend to present test data solely for simple regular motions observed in laboratory conditions. Unfortunately, vessels experience irregular and coupled motions at sea. While most sensors perform well in sine wave motion, only a select few excel in irregular sea conditions. Our MRUs stand out as they consistently deliver exceptional performance across all conditions.





Real Sea condtions

3rd party test facility	Motion Lab, University of Agder Grimstad (Norway)
Motion reference platform	Bosch Rexroth eMotion-8000 hexapod with laser position measurements
Tested MRU	MRU Marine 6000
Tested scenarios	Regular and irregular sea-state conditions Based on real sea conditions Single and multi-axis coupled motions Motion periods from 3 to 25 seconds



Lab conditions

This test verifies that the Norwegian Subsea MRUs meet the specifications and demonstrates typical MRU performance in realistic sea conditions.

Two Norwegian Subsea MRUs were mounted on top of a hexapod and the MRU output was compared to the reference measurements from the hexapod (truth).

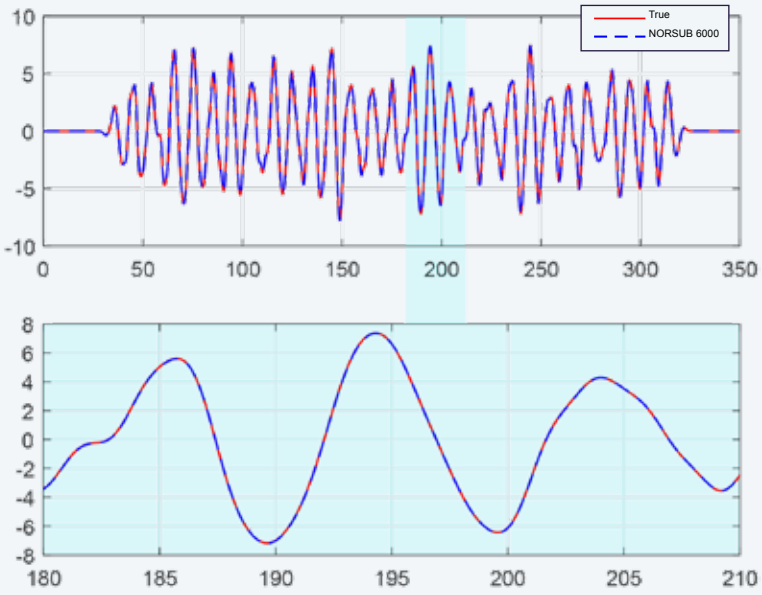
Both regular and irregular motion patterns were tested, with periods from 3 to 25 seconds and amplitudes from 0.1 to 0.8 m / 0 to 10 degrees. All degrees of freedom (DOFs) were tested individually and combined (coupled motions)

Combined roll and pitch test

Irregular motion profile with significant amplitude of 5 degrees and peak period of 10 seconds.

The test starts and stops with the MRU in the horizontal position (0 degrees).

The roll and pitch RMS error is calculated over the entire test duration of 350 seconds.



Roll performance

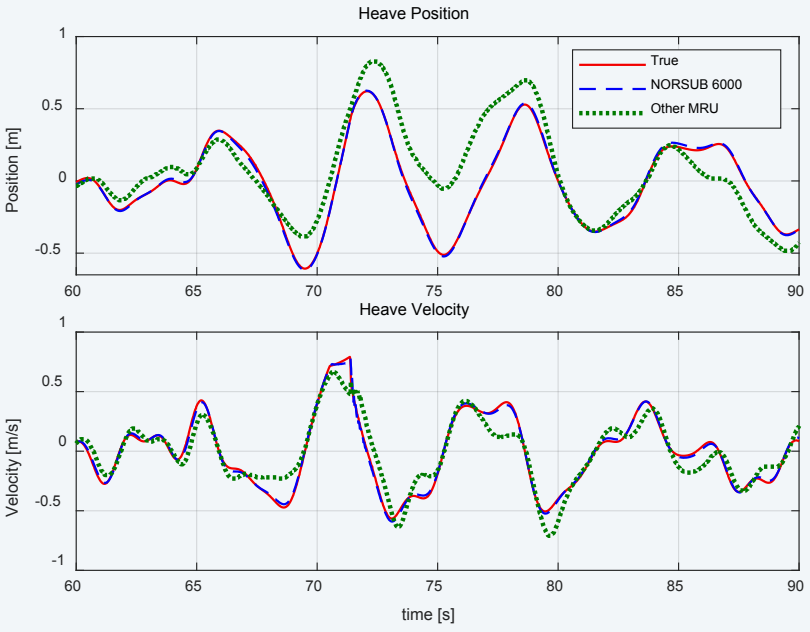
6000	*Datasheet Specs*	*Test Results*	*Approved*
Roll	0.02°	0.016°	✓
Pitch	0.02°	0.014°	✓

Heave Test

The Heave test has an irregular motion profile with a significant amplitude of 0.5 m and peak period of 7 seconds. The Heave filter uses automatic estimation of significant amplitude and peak period to optimize the performance.

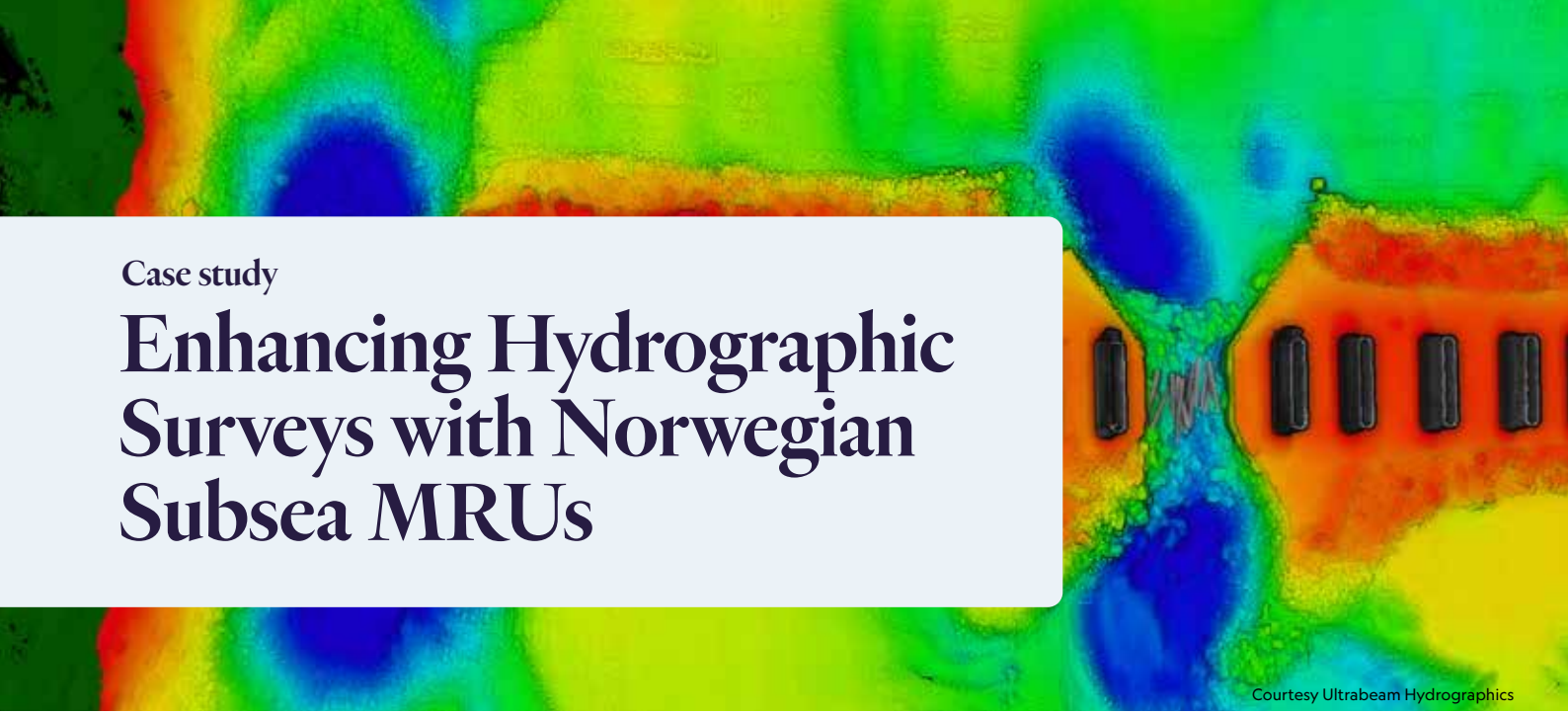
The Heave RMS error is calculated over the entire test duration of 250 seconds, with very accurate results.

For reference, an MRU from another brand, with comparable Heave accuracy specifications, was included in the tests. This MRU performed well in sine waves, but in real sea conditions, it was unable to provide useful results.



Heave position and velocity

6000	*Datasheet Specs*	*Test Results*	*Approved*
Heave	0.05 m or 5.0%	0.017 m	✓
Heave Velocity	0.02 m/s or 3.0%	0.011 m/s	✓



Courtesy Ultrabeam Hydrographics

Case study

Enhancing Hydrographic Surveys with Norwegian Subsea MRUs

Ultrabeam Hydrographic specializes in high-resolution 3D underwater surveys for marine asset inspections, including ports, bridges, and offshore wind farms. To ensure precise motion compensation, Ultrabeam integrated Norwegian Subsea's MRU 6000 Marine SW with their pole mounted Norbit Winghead, significantly improving survey accuracy and operational efficiency.

Ultrabeam

The Challenge: Size and Weight Constraints in Inland Surveys

Ultrabeam's Black Swan, a modular survey platform, conducts highly detailed 3D mapping, requiring precise motion data for roll, pitch, and heave compensation. However, conventional MRUs were bulky and cumbersome, making integration difficult—especially for inland and remote surveys where equipment weight is a critical factor.

"Every kilogram matters, and traditional MRUs were simply too heavy..."
– Gabriel Walton, Technical Director, Ultrabeam

The Solution: Norwegian Subsea's Compact MRUs

By adopting Norwegian Subsea's MRUs, Ultrabeam achieved seamless integration without compromising performance. The MRUs' compact size and calibration-free design reduced downtime, making surveys more efficient.

"We perform an initial calibration just once after installation. After that, there's no need for recalibration, significantly reducing our downtime."
– Gabriel Walton, Ultrabeam

The Impact: High-Quality Data and Increased Efficiency

Norwegian Subsea MRUs enabled Ultrabeam to collect accurate 3D survey data in challenging environments. The lightweight, cost-effective design improved logistics and ***reduced reliance on external RTK infrastructure***, expanding Ultrabeam's operational scope.

"While being the ideal companion in form and fit, the Norwegian Subsea MRU gives us motion compensation performance without needing external corrections"
– Gabriel Walton, Ultrabeam

With these enhancements, Ultrabeam continues to set new standards in hydrographic surveying, increasing efficiency and reliability in marine asset inspections.



Courtesy Wassp Ltd

Case study

Enhancing Multibeam Accuracy for Offshore Surveying and Fisheries

WASSP is a global provider of ***multibeam sonar technology***, delivering high-resolution seabed mapping solutions for ***commercial fishing, hydrographic surveying, and offshore industries.***

wassp® MULTIBEAM
SEE IT ALL

The Challenge: Overcoming RTK Limitations

WASSP sought an affordable yet precise solution for heave compensation without relying on expensive RTK (Real-Time Kinematic) positioning. RTK is widely used for high-precision surveying but is costly and unsuitable for deep-water environments. High-end Motion Reference Units (MRUs) offer an alternative, but their high price puts them out of reach for many users.

The Solution: Norwegian Subsea MRU

WASSP identified ***Norwegian Subsea's MRU*** as a highly accurate yet affordable option for motion compensation. After testing, it proved capable of delivering performance comparable to premium systems, making it an ideal solution for users operating beyond RTK coverage.

Testing & Validation

Extensive trials in ***New Zealand's challenging offshore conditions*** confirmed its reliability.

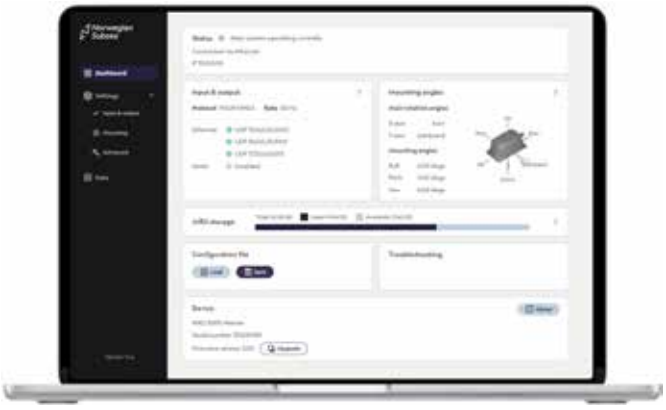
"In performance tests, we're getting incredibly close to RTK-level heave accuracy without the subscription costs or the burden of owning and maintaining a base station."

"For those who really want high-accuracy mapping that overlaps nicely and needs minimal cleaning, the better the sensor you use, the better the results you get. For us, the Norwegian Subsea MRU is now our premium offering. The price tag is much less than premium, however."
– Nick Fogarty, Senior Product Manager at WASSP

WASSP now offers the ***Norwegian Subsea MRU*** as a ***high-performance yet affordable motion compensation option*** for its customers.

High accuracy, easy to use and zero maintenance

User friendly configuration
All Norwegian Subsea MRUs and ACCs can be easily configured using the web-based configurator, where you can set communication ports, customise output protocols, adjust installation parameters, and check internal status. It can also update the MRU firmware, log the sensor data to file, and plot output data. Try the configurator yourself at demo.norwegian-subsea.no



Every sensor is calibrated, tested and validated
Every MRU is delivered with configuration, calibration, and validation certificates, and all MEMS sensors are temperature calibrated so sensor temperature does not influence the results. Every unit is calibrated and validated independently through a systematic sequence of rigorous tests in our labs.

No recalibration needed
The calibration certificate is valid for 4 years and full product specifications are maintained in this period under normal operating conditions. A slightly lower specification for Roll & Pitch is maintained during the lifetime of the sensor. For most applications, therefore no recalibration is required during the lifetime of the sensor.



Roll / pitch accuracy

	With recalibration	*Without recalibration*
3000	0.05°	0.1°
6000	0.02°	0.05°
9000	0.01°	0.02°

Retrofit of existing MRU
Many vessels have MRUs installed that need frequent recalibration and repair or are end-of-life since support and recalibration services are no longer available. Our MRUs use the latest generation of high-end MEMS sensors for extreme stability, making them ideal for retrofit installations with no need for future recalibration.

Typical retrofit applications
Any application that uses a motion reference unit should consider our retrofit solution. However, applications such as AHC, dynamic positioning and helideck motion monitoring can especially benefit as you can switch to an affordable MRU that does not require recalibration.

Export license
Our MRUs do not require an export license as they are not on the list of dual use items as per EU regulations.

About us



Norwegian Subsea is a leading provider of high-performance Motion Reference Units (MRU) and motion sensors for marine, subsea and offshore wind applications. Our products provide accurate and reliable measurements for control and monitoring purposes.

Since 2014, we've rapidly become a trusted global supplier of motion sensors. Today, over 1200 of our MRUs are in daily operation worldwide on over 600 vessels. Serving diverse industries such as ship motion monitoring, hydrography, green energy, and subsea oil production, our satisfied customers testify to the reliability and quality of our solutions.

Our mission is to develop exceptional and cost-effective motion sensors tailored for the marine, land, and subsea sectors. Through the fusion of advanced sensor algorithms, state-of-the-art MEMS sensors, and precision-engineered hardware, we create superior products. Rigorous testing in cutting-edge laboratories and real-world environments ensures the performance and dependability of our sensors.

With our headquarters situated in Oslo, Norway, Norwegian Subsea remains dedicated to providing innovative solutions that address the evolving needs of our customers in motion monitoring and beyond.

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Selected customer references

Over 1200 Norwegian Subsea MRUs are in operation world-wide on over 600 vessels, floating and subsea structures.

Hydrography

- Our MRUs are used with the following sonar systems

GeoAcoustics
R2SONIC
NORBIT
TELEDYNE RESON
wassp MULTIBEAM SEE IT ALL

Dynamic positioning

- Our MRUs are used with the following DP systems

BRUNVOLL
MT
KONGSBERG

Wave radar

MIROS
RADAC
RS AQUA

Helideck Monitoring Systems

ShoreConnection
SWIRE
OBSERVATOR group
wise GROUP
nevis TECHNOLOGY
VAISALA

Floating wind lidar buoys

비갠
Blue Aspirations
CLS COLLECTE LOCALISATION SATELLITES
FUGRO
GEPS TECHNO
SEATECH
WOODS HOLE GROUP A CLS COMPANY

Motion Compensation of cranes, gangways, LARS & winches

Allseas
AXTech
Cranemaster
ENTEC
Hydrauvision Driven by technology
MOTUS
OCEANEERING
PALFINGER
SH GROUP
SMST
safeLINK
SEAONICS
SEAQUALIZE
SERVI GROUP
UPTIME
VHT VAN HALTEREN TECHNOLOGIES
Zbridge Offshore Access Solutions
ULMATEC

Other references

ARGUS
Baker Hughes
National Research Council of Italy
core marine
FORCE TECHNOLOGY
HITEC PRODUCTS
HÖEGH LNG
IHC
Jan De Nul GROUP
LIGHT STRUCTURES
NIOT
pliant OFFSHORE
RN
SAIPEM
SINTEF
SUBSEA CONNECT
OSM Thome
YOKOGAWA



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